

Monsoon, 2024

B. Tech 1st Semester Examination

Design Thinking & Idea Lab – Practical

Course Code: BME23103P

Marks – 50

Time

*The figure in the margin indicates full marks for the questions.***SECTION 1**

ALL QUESTIONS ARE COMPULSORY

- (a) Which of the following is a subtractive prototyping process
- (a) Five axis CNC milling
 - (b) Fused Deposition Modelling
 - (c) Multi Jet Modelling
 - (d) Stereolithography
- (b) Which of the following terms is not related to additive manufacturing
- (a) Layer by Layer
 - (b) Physical model
 - (c) 3D CAD data
 - (d) Production line
- (c) Additive Manufacturing is a term which refers to fabrication technology material layer by layer to an existing substrate
- (a) TRUE
 - (b) FALSE

Which of the following is the Pre-processing stage?

- (a) Remove support
- (b) Checking 3D CAD data
- (c) De-powdering loose material
- (d) Dip in binder to strengthen part

What file format do we export to print

- (a) OBJ
- (b) STL
- (c) PRT
- (d) IGS

Which of the following is NOT a 3D printing technology

- (a) FDM

- (b) ABS
 - (c) SLA
 - (d) SLS
- (g) The surface on which prints are built is
- (a) Build plate
 - (b) Extruder
 - (c) Runout sensor
 - (d) Slicer
- (h) 3D printing can be performed using stainless steel as printing material
- (a) TRUE
 - (b) FALSE
- (i) Which of the following sectors can use 3D Printing
- (a) Aerospace engineering
 - (b) Automobile engineering
 - (c) Reverse engineering
 - (d) All of these
- (j) Which of the following is not used in 3D printing
- (a) ABS
 - (b) Nylon
 - (c) PLA
 - (d) PVC

SECTION 2

ALL QUESTIONS ARE COMPULSORY

Discuss the colour code of different resistors with examples.

Why do 3D printed parts show "Stair Stepping" effect?

Explain various electronic components you have work in the laboratory.

SECTION 3

ALL QUESTIONS ARE COMPULSORY

Explain the working of moisture sensor? Discuss the procedure of imp moisture sensor-based circuit.